

Concepts of Operating Systems And Linux Operating Systems

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Linux File system Structure:-

- **While an installation of Linux system we need to create two partitions:**
  - **swap partition (will be use by Linux as a swap area)**
  - **"/" (root) partition also called as data partition**
- Linux follows FHS i.e. **File System Hierarchy Standards**
- Data can be store into the "**data partition**" in an organized manner, so linux filesystem structure is having **hierarchical structure**, which is like an **inverted tree.**
- When any user logged in into the system or **opens a terminal, system takes the user by default into the user's home dir.**
- **Linux file system structure starts with root dir ("/"),** and all the data can be stored or kept inside it **under sub-dir's in an organized manner.**
- **root** dir i.e. "/" dir contains sub-dir's like: "/**boot**", "/**bin**", "/**sbin**", "/**etc**","/**lib**", "/**usr**", "/**home**", "/**root**" "/**mnt**", "/**dev**" etc....

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Linux File system Structure:

- **“/boot” :**
  - It contains information about booting the system
  - It contains Linux kernel by the name **vmlinuz.**
  - **grub** - boot loader
  - **config** - kernel configuration
  - **initrd/initramfs** - initail root file system
- **"/bin” :** contains user commands in a binary format, e.g. commands like ls, cat, cp, mv ,etc...
- **"/sbin” :** contains admin/system commands in a binary format, e.g. lscpu, adduser, deluser, mkfs etc...
- **“/lib”** – Contains essential shared libraries (.so) and kernel modules (.ko) required for system boot and basic programs.
- **"/etc“ :** contains host specific system configuration files.
  - In computing configuration files (or config files) are files used to configure the parameters and initial settings for some computer programs.

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Linux File system Structure

- **/dev** – Contains device files for all hardware devices (e.g. char and block devices).
- **/mnt** – Temporary mount point for mounting other file systems manually.
- **/opt** – Contains optional or third-party software packages.
- **/proc** – It providing information about system processes and kernel.
- **/sys** – It used for device management and kernel settings.
- **/tmp** – Temporary files; contents are usually cleared on reboot.
- **/usr** – Contains user-installed software and shared read-only data (includes binaries, libraries, docs, etc.).
- **"/root“ :** root is a home directory of root user.
- **"/home” :** contains home dir's of all users for multi-user system.
  - For any user on its creation of an account by default sub directory by the name of the user got created by the operating system, in which that user can store data, user can have read, write as well execute perms in that directory .

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Path

- It is an **unique location of any file in a file system structure that can be represented in a string format** .
- It is sequence of chars format separated filenames /sub-dir names by **delimiter char or delimiter.**
- There **are three delimiter chars**:
  1. **slash (/) -- In Linux**
  2. **backslash (\ ) -- In Windows**
  3. **colon (:) -- In Linux as well as in Windows**
- **In Linux there are two ways by which we can mention path of any file/dir:**
  1. **Absolute path:** It is a full path of any file/dir which starts with root dir ("/").
     - Path which starts with “/” is called as absolute path.
     - E.g. /home/sunbeam/MyData/Demos/demo01.sh
  2. **Relative path:** Path with respect to current directory is called as relative path
     - E.g. MyData/Assignments/assign02.pdf .

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File Management

Q. What is file?

- **User view:**
  - File is a named collection of logically related information/data.
  - File is a container which contains logically related information/data.
  - File is a collection of characters/records/lines.
  - File is a basic storage unit
- **System view:**
  - File is a **stream of bits/bytes.**
    - **File = data + metadata**
      - data = actual file contents
      - metadata = information about the file.

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- Data of the file is exists inside the file whereas information about the file gets stored inside one structure referred as **FCB (File Control Block).**
- In UNIX environment FCB is also called as an **iNode**.
- **FCB / iNode contains information about the file like:**
  1. types of files
  2. size: number of bytes
  3. links: number of hard links
  4. mode (permissions): (u) rwx, (g) rwx, (o) rwx
  5. user & group
  6. time-stamps: modification, creation, access.
  7. info about data blocks
- terminal> ls -l
  - type, mode, links, user, group, size, timestamp, name.
- terminal> stat filepath

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❖**In Linux / UNIX based OS's there are 7 types of files:**

1. **regular file (-)** : all text files, source files, image files, audio files, video files, executable files etc..
2. **directory file (d) :** it is a special type of file in UNIX which is a like a container whose contents are names of files and sub directory in it.
3. **character special device file (c) :**
4. **block special devices file (b) :**
5. **linkable file (l)** : special type of file in Linux which is same like short cut files in Windows, which contains link of an original file and by using this linkable file contents of original file also can be accessed.
   - To access deeply located file more frequently we should create its linkable file.
6. **pipe file (p) :**
7. **socket file (s) :**

- **Pipe file and socket file use in IPC.**

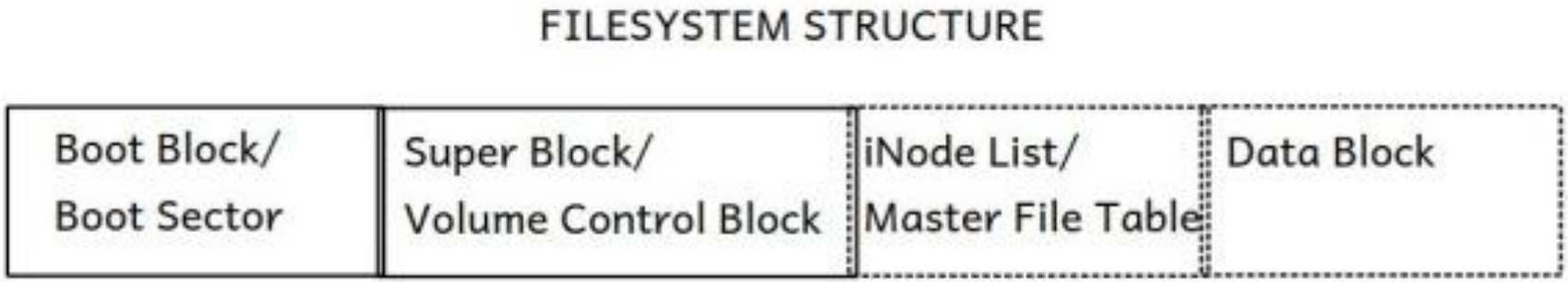
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Operating Systems Concepts

- Per file one iNode/FCB gets created by the system and **i.e**. **hence no. of iNodes = no. of files onto the disk.**
- **data + metadata** of all the files are kept onto the disk, as disk may contents thousands of files, so thousands of iNodes and millions of bytes of data gets stored onto the disk, and hence all this **data + metadata** of all files need be keep onto the disk in an organized manner so that it can be accessed efficiently.
- **File system :** file system is a way to store data onto the disk in an organized manner so that it can be accessed efficiently and conveniently.
  - e.g. Each OS has its own file system like,
  - UNIX: UFS(UNIX File system),
  - Linux: Extended File system ext2, ext3, ext4,
  - Windows: FAT, NTFS etc...,
  - MAC OSX: ReiserFS, XFS, HFS(Hierarchical File system) etc...

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- **Files system Structure:** File system divides disk/partition logically into sectors/blocks, like boot sector/boot block, volume control block/super block, master file table/iNode list block and Data Block.



1. Boot Block: It contains information about booting the system like bootstrap program, bootloader etc...
2. Super Block: It contains information about remaining sections, like total no. of data blocks, no. of free data blocks, no. of allocated data blocks etc....
3. iNode List: It contains linked list of iNode's of all files exists on a disk.
4. Data Block: It contains actual data.

- **File System logically divide partition into 4 sections.**
  - **Boot block/Boot sector**
    - Contains programs/info required for booting of OS
    - Typically contains bootstrap program and bootloader program
  - **Super block/Volume control block**
    - Contains information of whole partition.
    - Capacity, Label.
    - terminal> df -h
      - Total number of data blocks/inodes.
      - Number of used/free data blocks/inodes.
      - Information of free data blocks/inodes.
  - **Inode List/Master file table**
    - Inodes (FCB) for each file
  - **Data blocks**
    - Stores data of the file.
    - Each file have zero or more data blocks.
    - Size of data blocks can be configured while creating file system
    - File system is created by the format utility while formatting the partition.

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Creating File Systems in Windows and Linux

- **Windows:**
  - format.exe - : utility used to format a partition.
- **Linux:**
  - mkfs - : make file system command.
    - **Example:**
    - $ sudo mkfs -t ext3 /dev/sdb1
    - $ sudo mkfs -t vfat /dev/sdb1
      - -t <type> : specifies the file system type.
      - Common types : ext3, ext4, vfat, ntfs, xfs, etc.
      - Example partition name : /dev/sdb1

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Directory File

- **From an end-user perspective, a directory is a container that holds files and sub-directories.**
- **From the OS perspective, a directory is treated as a special file.**
- **Internal Structure**
  - A directory file contains entries for each file/sub-directory.
  - Each entry stores: i-node number and Name of file or sub-directory
  - **Example Command**
    - **terminal> ls -a -i -1 /home/sunbeam**
    - -a → show hidden files
    - -i → display inode numbers
    - -1 → one entry per line
- **Directory Listing**
  - **terminal> ls dirpath**
  - Lists contents of a directory

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Directory

- **Directory Permissions / Modes**
  - **Read Permission (r)**
    - Allows listing of directory contents
    - Enables reading directory entries (names of files/sub-directories)
    - Example: ls command works
  - **Write Permission (w)**
    - Allows modifying directory contents
    - Create new files and sub-directories
    - Delete files or sub-directories
    - Rename files
  - **Execute Permission (x)**
    - Allows access/traversal of the directory
    - Enables entering the directory using cd
    - Required to access files inside the directory
  - **Directory permissions apply to the directory itself, not the files inside it**
  - **Example:**
    - Without x → Cannot enter directory (even if r is given)
    - Without r → Cannot list files (even if inside directory)

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Link

- **Links are shortcuts to access deeply located files quickly.**
- **Hard Link**
  - A **hard link** is an **additional name** for an existing file. It points directly to the **same inode** (data on disk), so multiple filenames can reference the exact same data.
  - This way the hard link gets all the attributes of the original file and points to the same data block as the original file.
  - **Terminal : $ln [source_file] [link_name]**

  - **A new directory entry is created, which has a new name and same inode number.**
  - **No new file (inode and data blocks) is created.**
  - **Link count in the inode of the file is incremented.**
  - **If directory entry of target file is deleted (rm command), file can be still accessed by link directory entry.**
  - **Cannot create hard link for directories, because it may lead to infinite recursion (while traversing directories recursively e.g. ls -R)**

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Link File

- **Symbolic Link**
  - **A symbolic link, also known as a symlink or soft link, is a special type of file that points to another file or directory.**
  - **Terminal : $ln -s [target_file_or_dir] [symlink_name]**
- **symlink()**
  - **syscall A new link file is created (new inode and new data block is allocated), which contains info about the target file (absolute or relative path).**
  - **Link count is not incremented.**
  - **If target file is deleted, the link becomes useless.**
  - **Can create symlinks for directories as well as file also.**

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Thank you!

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